

SUBHRANEEL MUKHOPADHYAYA

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EDUCATION

University of Southern California Los Angeles, CA
Master's in Computer Science August 2025-May 2027

- Relevant Coursework : Analysis of Algorithms, Information Retrieval

New Horizon College of Engineering Bangalore, India
Bachelor's in Computer Science and Engineering November 2021-August 2025
GPA:9.34/10

- First Class with distinction
- Relevant Course work : Data mining and Machine Learning, Linux System Programming, Cloud Computing
- Student Clubs : Innovation Club

TECHNICAL SKILLS

Languages: Python, SQL, C++
Cloud Computing: GCP VM Management, GCP Bucket Data Management, Cloud Storage pipelines (GCS), Vertex AI integration
Software Development: Data Structures & Algorithms, REST API Development (Flask/FastAPI), Linux/SSH
Machine Learning: Proximal Policy Optimization, Artificial Neural Networks (TensorFlow, Keras), NLP (Sentence Transformers, LoRA/QLoRA, RAG), Statistical Machine Learning
Developer Tools: GitHub, hydra configurations, Conda & Virtual Environments, Docker

EXPERIENCE

Google Cloud x Hack2Skill Agentic AI Hackathon Bangalore, India
Command Center Volunteer July 2025-July 2025

- Ran operations during a 30-hour event (participant support, schedule control, incident triage, judge/mentor coordination) using google sheets and company made apps delivering a record-setting hackathon experience

Tejas Networks Bangalore, India
R&D Intern February 2025-May 2025

- Configured O-RAN to work in a virtualized environment to monitor different KPIs (Buffer Backlog, Throughput, stalls) between user equipments for PPO scheduling experiments and achieved a 40% increase in throughput
- Modified SRSran to calculate latency between two UEs and created a pipeline to transport it back to RIC via zeroMQ to leverage latency as a metric for the PPO algorithm

Lakshya Space Bangalore, India
Intern September 2023-October 2023

- Built an end-to-end ML notebook on a sample engine-pressure dataset, cleaned data with Pandas/NumPy and explored trends with Matplotlib
- Evaluated models with RMSE, MAE, and R, selected final model based on validation RMSE and stability across runs

PROJECTS

SciBot - Science Research assistant September 2025-September 2025

- Automated a web app to facilitate research in science. The web app scrapes open-access articles from PLOS, stores in GCS, indexes with FAISS, and uses SentenceTransformers for embeddings
- User queries retrieve top k ranked article links from FAISS, the top 3 articles are downloaded from GCS bucket and summarized with Vertex AI, optimized to the user's query and profession

Research paper on AI-Driven Indian Language translation December 2024-June 2025

- Co-authored a research paper on Future Directions in AI-Driven Indian Language Translation presented at the International Conference on TechWork (TETWT-2025)
- Explored different methods in multilingual translation, benchmarked MuRIL against mBERT on XTREME tasks and proposing a hybrid RNN-based approach for low-resource Indian languages

Video Audio translator November 2024-December 2024

- Fine-tuned Mistral-7B with LoRA for video subtitle translation and published adapters on Hugging Face, the LoRA model underperformed with respect to zero-shot Mistral on BLEU
- shipped a zero-shot pipeline (ASR translation subtitle sync) for Hindi/Kannada/Telugu, achieving >85% accuracy with <500 ms end-to-end latency